

Project Executable Files

Date	5th July 2026
Team ID	RAHMA HUSSEIN JUMA
Project Name	CREDIT CARD APPROVAL REQUEST
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes — schema.sql
5	Environment configuration file or (.env.example similar)	Yes — .env.example
6	Deployed application URL (if hosted)	No — runs locally only; Watson ML deployment not completed
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA — this is a web app
8	Dockerfile / Containerization config (if applicable)	No

9	Test files and test results	Yes — load_test.py, model_comparison_results.csv, Project Testing doc
10	Demo video or walkthrough (if required)	Yes: https://youtu.be/Re5yA01it_0

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

/Credit-Card-Approval-Prediction/

- |— README.md
- |---schema and env
 - |— schema.sql
 - |— .env.example
- |— /1. Brainstorming & Ideation
 - |--Brainstorming & ideation
 - |--Define problem statement
 - |--Empathy map
- |— /2. Requirement Analysis
 - |--Data flow diagram
 - |--Solution requirements
 - |--Technology stack
- |— /3. Project Design Phase
 - |--Problem solution fit
 - |--Proposed solution
 - |--Solution architecture
- |— /4. Project Planning Phase
 - |--Project planning
- |— /5. Project Development Phase
 - |--Code-layout, readability and usability
 - |--Coding & solution
 - |--No of functional features included in the solution
- |— /6. Project Testing
 - |--Performance testing
 - |--load test.py
- |— /7. Project Documentation
 - |--Final report
 - |-- Project_Documentation
 - |--project demo video
 - |--Project executable files
- |— /9.Source Code Files
 - |-- dataset
 - |--application_record.zip
 - |--Credit_record.csv
 - |— /epic2_visualizing_analysing
 - |--charts
 - |--epic2_eda(1).py
 - |— /epic3_preprocessing
 - |--epic3_preprocessing.py
 - |--processed_data.csv
 - |— /epic4_model_building

```

|--epic4_model_building.py
|--model_comparison_results.csv
└─ /epic5_application_building
    |--Model
        |--best_model.pkl
        |--model_columns.pkl
        |--scaler.pkl
    |--Static
        |--Style.css
    |--templates
        |--index.html
        |--result.html
    |--app.py
    |--load_test.py
    |--requirements.txt
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not deployed-local only
Login Credentials (Demo)	Not applicable — no authentication implemented
Platform / Hosting Provider	Local (Flask dev server); IBM Watson ML planned for future cloud hosting
Repository Link	https://github.com/rahmahusseinjuma-beep/Credit-Card-Approval-Prediction
Demo Video Link	https://youtu.be/Re5yA01it_0

Step 4: Run Instructions

Step 1: Clone the repository:

```

git clone < https://github.com/rahmahusseinjuma-beep/Credit-Card-Approval-Prediction
>
  
```

Step 2: Navigate to the application folder:

```

cd "9.Source code files( Project Development Phase)/epic5_application_building"
  
```

Step 3: Install dependencies:
pip install -r requirements.txt

Step 4: Run the application:
python app.py

Step 5: Open a browser at:
<http://127.0.0.1:5000>

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Low precision on risky-applicant predictions, caused by severe class imbalance in the training data (only 1.7% of applicants are labeled risky)	Mitigated with <code>class_weight="balanced"</code> during training; full fix (e.g. SMOTE) listed under Future Scope
2	No error handling for invalid or missing form input	Not yet implemented; noted for future improvement
3	Application runs on Flask's development server only, not production-hosted	IBM Watson Machine Learning deployment planned but not completed